

A distortion < 2 grid obstruction for coarse-Lipschitz c_0 embeddings into duals

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Source and Target

The source paper is Braga–Lancien–Petitjean–Prochazka’s arXiv:1909.12132 paper [1]. The paper asks whether c_0 coarsely Lipschitz embeds into a separable dual space. Its Theorem 6.2 proves a quantitative obstruction: if c_0 coarse-Lipschitz embeds into X^* with distortion strictly smaller than $3/2$, then X contains ℓ_1 . Immediately afterward, the paper obtains a distortion < 2 obstruction for an auxiliary graph M , while noting that it is not clear whether the graph M is isometric to a subset of c_0 .

The counterpart is that one has to define by hand the required topology τ and then check that the distance is τ -lower semicontinuous (which was automatic with the w^* -topology in ℓ_∞). The same optimal estimate on the Szlenk index is obtained.

6. LOW DISTORTION EMBEDDING OF THE GRID OF c_0 INTO DUALS

In this section, we produce two uniformly discrete countable metric spaces so that if they Lipschitz embeds into X^* with Lipschitz distortion strictly less than $3/2$ or 2 , respectively, then X must contain an isomorphic copy of ℓ_1 . We use this in order to prove Theorem 6.2.

We define the *integer grid* of c_0 as

$$\text{Grid}(c_0) = \{(x_n)_n \in c_0 : \forall n \in \mathbb{N}, x_n \in \mathbb{Z}\}.$$

So $\text{Grid}(c_0)$ is a $(1, 1)$ -net of c_0 (meaning that it is 1-separated and for every $x \in c_0$, $d(x, \text{Grid}(c_0)) \leq 1$). We consider it as a metric space with the metric inherited from c_0 .

Proposition 6.1. *Let X be a Banach space and $f : \text{Grid}(c_0) \cap 2B_{c_0} \rightarrow X^*$ be a Lipschitz embedding with distortion strictly smaller than $\frac{3}{2}$. Then X contains an isomorphic copy of ℓ_1 .*

Proof. Replacing f by λf for some appropriate $\lambda > 0$, we may assume that there exists $D \in [1, \frac{3}{2})$ so that

$$\|x - y\| \leq \|f(x) - f(y)\| \leq D\|x - y\|$$

for all $x, y \in \text{Grid}(c_0) \cap 2B_{c_0}$. Fix $\varepsilon > 0$ such that $3 - \varepsilon - 2D > 0$.

Let $(e_n)_n$ be the canonical basis of c_0 . For every $k \in \mathbb{N}$, pick $x_k \in S_X$ such that

$$\langle x_k, f(2e_k) - f(-e_k) \rangle \geq 3 - \varepsilon.$$

We claim that the sequence $(x_k)_k$ has no weakly Cauchy subsequence. Indeed, let $M = \{m_1 < m_2 < \dots\} \in [\mathbb{N}]^\omega$ and set $A_1 = \{m_{2k+1} : k \in \mathbb{N}\}$ and $A_2 = \mathbb{N} \setminus A_1$. Then, for all $k \in \mathbb{N}$ and all $m > m_{2k+1}$ we have that

$$\begin{aligned} \langle x_{m_{2k+1}}, f(\mathbf{1}_{A_1 \cap [1, m]}) - f(\mathbf{1}_{A_2 \cap [1, m]}) \rangle &= \langle x_{m_{2k+1}}, f(2e_{m_{2k+1}}) - f(-e_{m_{2k+1}}) \\ &\quad + f(\mathbf{1}_{A_1 \cap [1, m]}) - f(2e_{m_{2k+1}}) \\ &\quad + f(-e_{m_{2k+1}}) - f(\mathbf{1}_{A_2 \cap [1, m]}) \rangle \\ &\geq 3 - \varepsilon - 2D \end{aligned}$$

and

$$\begin{aligned} \langle x_{m_{2k}}, f(\mathbf{1}_{A_1 \cap [1, m]}) - f(\mathbf{1}_{A_2 \cap [1, m]}) \rangle &= \langle x_{m_{2k}}, f(-e_{m_{2k}}) - f(2e_{m_{2k}}) \\ &\quad + f(\mathbf{1}_{A_1 \cap [1, m]}) - f(-e_{m_{2k}}) \\ &\quad + f(2e_{m_{2k}}) - f(\mathbf{1}_{A_2 \cap [1, m]}) \rangle \\ &\leq -3 + \varepsilon + 2D. \end{aligned}$$

Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ and set

$$x^* = w^*\text{-}\lim_{m, \mathcal{U}} (f(\mathbf{1}_{A_1 \cap [1, m]}) - f(\mathbf{1}_{A_2 \cap [1, m]})).$$

The above inequalities imply that for all $k \in \mathbb{N}$:

$$x^*(x_{m_{2k+1}}) \geq 3 - \varepsilon - 2D \quad \text{and} \quad x^*(x_{m_{2k}}) \leq -3 + \varepsilon + 2D.$$

This shows that $(x_{m_k})_k$ is not weakly Cauchy.

By Rosenthal's ℓ_1 theorem [31], this implies that $(x_k)_k$ has a subsequence equivalent to the standard unit basis of ℓ_1 . In particular, X contains an isomorphic copy of ℓ_1 and we are done. $\square$

Theorem 6.2. *If c_0 coarse Lipschitz embeds into a dual space X^* with coarse Lipschitz distortion strictly less than $\frac{3}{2}$, then X contains an isomorphic copy of ℓ_1 .*

Proof. Assume f is such a coarse Lipschitz embedding from c_0 into X^* . Replacing f with $x \mapsto f(nx)/n$ for a large enough $n \in \mathbb{N}$, the map f restricted to $\text{Grid}(c_0) \cap 2B_{c_0}$ becomes a Lipschitz embedding with distortion strictly smaller than $3/2$. Then, it follows from Proposition 6.1 that X contains an isomorphic copy of ℓ_1 . $\square$

We will now show that replacing $\text{Grid}(c_0) \cap 2B_{c_0}$ by an appropriate graph M , the distortion in Proposition 6.1 can be pushed up to 2. It is not clear if the metric space M needed for this is isometric to a subset of c_0 .

We define the graph M as follows. Let $S = [\mathbb{N}]^{<\omega}$, $G = \mathbb{N}$ and $H = \mathbb{N}$. Moreover, we write $G = \{g_i : i \in \mathbb{N}\}$ and $H = \{h_i : i \in \mathbb{N}\}$, where $g_i = h_i = i$ for all $i \in \mathbb{N}$. We define M as the disjoint union

$$M = \{0\} \sqcup S \sqcup G \sqcup H$$

and we define a graph structure on M by putting an edge

- between 0 and any element of S ,
- between $A \in S$ and $g_k \in G$ iff $k \in A$, and
- between $A \in S$ and $h_k \in H$ iff $k \notin A$.

Then endow M with the shortest path distance, which we denote by d_M . It should be clear that $d_M(A, B) = 2$ if $A \neq B \in S$ and that $d_M(g_k, h_k) = 4$ for all $k \in \mathbb{N}$.

Proposition 6.3. *Let $f : (M, d_M) \rightarrow X^*$ be a Lipschitz embedding with distortion strictly less than 2. Then X contains ℓ_1 .*

This proposition should be compared with [29, Theorem 3.1] where a metric space (N, d) is constructed such that if N Lipschitz embeds into X with distortion strictly less than 2 then X contains ℓ_1 .

Proof. Replacing f by λf for some appropriate $\lambda > 0$, we may assume that there exists $D \in [1, 2)$ so that

$$\|x - y\| \leq \|f(x) - f(y)\| \leq D\|x - y\|$$

for all $x, y \in M$. Fix $\varepsilon > 0$ such that $4 - \varepsilon - 2D > 0$.

For every $k \in \mathbb{N}$, let $x_k \in S_X$ be such that $\langle x_k, f(g_k) - f(h_k) \rangle \geq 4 - \varepsilon$. We claim that $(x_k)_k$ does not contain any weakly Cauchy subsequence. Rosenthal's ℓ_1 theorem thus implies that X contains ℓ_1 . In order to prove our claim, let $\mathbb{M} = \{m_1 < m_2 < \dots\}$ be an infinite subset of $\mathbb{N}$. We set $A = \{m_{2k+1} : k \in \mathbb{N}\}$ and $B = \mathbb{N} \setminus A$. Let $\mathcal{U}$ be a nonprincipal ultrafilter on

Result

The $3/2$ threshold in the source's grid argument can be replaced by 2. No auxiliary graph is needed.

Theorem 1. *Let X be a Banach space and let*

$$f : \text{Grid}(c_0) \cap 2B_{c_0} \longrightarrow X^*$$

be a Lipschitz embedding with distortion strictly smaller than 2. Then X contains an isomorphic copy of ℓ_1 .

Proof. After multiplying f by a scalar, choose $D < 2$ such that

$$\|u - v\|_\infty \leq \|f(u) - f(v)\| \leq D\|u - v\|_\infty$$

for all $u, v \in \text{Grid}(c_0) \cap 2B_{c_0}$. Pick $\varepsilon > 0$ with

$$4 - \varepsilon - 2D > 0.$$

Let (e_k) be the canonical basis of c_0 . For each $k \in \mathbb{N}$, choose $x_k \in S_X$ such that

$$\langle x_k, f(2e_k) - f(-2e_k) \rangle \geq 4 - \varepsilon.$$

We prove that (x_k) has no weakly Cauchy subsequence.

Let $M = \{m_1 < m_2 < \dots\}$ be an infinite subset of $\mathbb{N}$. Put

$$A = \{m_{2j+1} : j \in \mathbb{N}\}.$$

For $m \in \mathbb{N}$, define the signed truncation

$$s_A^m = \sum_{i \in A \cap [1, m]} e_i - \sum_{i \in [1, m] \setminus A} e_i$$

and similarly $s_{\mathbb{N} \setminus A}^m$. These vectors belong to $\text{Grid}(c_0) \cap B_{c_0}$.

If $k = m_{2j+1} \in A$ and $m \geq k$, then

$$\|s_A^m - 2e_k\|_\infty = 1 \quad \text{and} \quad \|s_{\mathbb{N} \setminus A}^m + 2e_k\|_\infty = 1.$$

Therefore

$$\begin{aligned} & \langle x_k, f(s_A^m) - f(s_{\mathbb{N} \setminus A}^m) \rangle \\ & \geq \langle x_k, f(2e_k) - f(-2e_k) \rangle - \|f(s_A^m) - f(2e_k)\| - \|f(s_{\mathbb{N} \setminus A}^m) - f(-2e_k)\| \\ & \geq 4 - \varepsilon - 2D. \end{aligned}$$

If $k = m_{2j} \notin A$ and $m \geq k$, the same calculation with signs reversed gives

$$\langle x_k, f(s_A^m) - f(s_{\mathbb{N} \setminus A}^m) \rangle \leq -4 + \varepsilon + 2D.$$

Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$, and set

$$x^* = w^* - \lim_{m, \mathcal{U}} (f(s_A^m) - f(s_{\mathbb{N} \setminus A}^m)).$$

The vectors inside the limit are norm bounded, so this weak-star ultralimit exists in X^* . The preceding inequalities imply that, for every j ,

$$x^*(x_{m_{2j+1}}) \geq 4 - \varepsilon - 2D \quad \text{and} \quad x^*(x_{m_{2j}}) \leq -4 + \varepsilon + 2D.$$

Thus (x_{m_j}) is not weakly Cauchy. Since M was arbitrary, (x_k) has no weakly Cauchy subsequence. By Rosenthal's ℓ_1 theorem, X contains an isomorphic copy of ℓ_1 . $\square$

Theorem 2. *If c_0 coarse-Lipschitz embeds into a dual space X^* with coarse-Lipschitz distortion strictly smaller than 2, then X contains an isomorphic copy of ℓ_1 .*

Proof. Let $F : c_0 \rightarrow X^*$ be such an embedding. There are constants $A, B, C, E > 0$, with $AC < 2$, such that

$$\frac{1}{A} \|x - y\|_\infty - B \leq \|F(x) - F(y)\| \leq C \|x - y\|_\infty + E$$

for all $x, y \in c_0$.

For $n \in \mathbb{N}$, define $F_n(x) = F(nx)/n$ on $\text{Grid}(c_0) \cap 2B_{c_0}$. If $u \neq v$ in this grid, then $\|u - v\|_\infty \geq 1$, and hence

$$\left(\frac{1}{A} - \frac{B}{n}\right) \|u - v\|_\infty \leq \|F_n(u) - F_n(v)\| \leq \left(C + \frac{E}{n}\right) \|u - v\|_\infty.$$

For n large enough, the lower coefficient is positive and

$$\frac{C + E/n}{1/A - B/n} < 2.$$

Thus F_n is a Lipschitz embedding of the grid with distortion strictly smaller than 2. Theorem 1 gives $\ell_1 \subset X$. $\square$

Corollary 1. *The space c_0 does not coarse-Lipschitz embed with distortion strictly smaller than 2 into any separable dual Banach space.*

Proof. If X^* is separable and X contained an isomorphic copy of ℓ_1 , then the adjoint of the embedding $\ell_1 \hookrightarrow X$ would give a bounded quotient map from X^* onto ℓ_∞ , impossible because a quotient of a separable Banach space is separable. Apply Theorem 2. $\square$

Verification Notes

The only new ingredient relative to Proposition 6.1 of the source paper is the signed truncation s_A^m . For $k \in A$, it is at distance 1 from $2e_k$, while $s_{\mathbb{N} \setminus A}^m$ is at distance 1 from $-2e_k$. This keeps the two Lipschitz-error terms equal to D , but the main separation is $\|2e_k - (-2e_k)\|_\infty = 4$ rather than 3. The Rosenthal theorem step and the coarse scaling step are otherwise exactly the same as in the source.

The result is partial. It does not answer whether c_0 coarse-Lipschitz embeds into a separable dual with distortion at least 2, nor whether c_0 coarsely embeds into a separable dual without a distortion bound.

Novelty and Literature Check

The local run indexes were searched for arXiv:1909.12132 and for terms around c_0 , separable duals, Kalton interlaced graphs, and coarse-Lipschitz distortion. No prior local packet for this arXiv id was found. Web/arXiv searches on 2026-06-28 for matching phrases around c_0 , separable duals, and distortion 2 surfaced the source paper but no exact duplicate of this signed-grid improvement.

Human Review Recommendation

Check the signed-truncation replacement in the proof of Proposition 6.1 and the passage from coarse-Lipschitz distortion < 2 to a Lipschitz embedding of the bounded integer grid. Those are the only points where the packet deviates from the source proof.

References

- [1] B. de Mendonca Braga, G. Lancien, C. Petitjean, and A. Prochazka. *On Kalton's interlaced graphs and nonlinear embeddings into dual Banach spaces*. arXiv:1909.12132, 2019.